

# Harry Lynch

Long Branch, NJ | 732-861-4011 | [hlynch02@tufts.edu](mailto:hlynch02@tufts.edu) | [LinkedIn](#) | [GitHub](#) | [harrylynch.dev](http://harrylynch.dev)

## EDUCATION

### Tufts University

Medford, MA

*B.S. Computer Science, Minor in Mathematics | GPA: 3.8/4.0 (Major: 3.9)*

*Expected May 2027*

- **CS Coursework:** Internet-Scale Distributed Systems, Parallel Computing, Networks, Database Systems, AI
- **Math Coursework:** Probability, Linear Algebra, Multivariable Calculus, Discrete Math, Abstract Algebra

## EXPERIENCE

### Bloomberg LP

New York, NY

*Software Engineer Intern – Buy-Side Compliance, Enterprise Products*

*June 2026 – August 2026*

- Owning the end-to-end rewrite of the **C++23** backend serving **VMGR**, Bloomberg's buy-side **compliance-violation monitoring** terminal function used by traders and auditors at some of the largest asset managers, hedge funds, and pension funds
- Moving all filtering, sorting, and pagination of **1M+ row** violation datasets server-side with **infinite scroll**, streaming **real-time** violation rows off a message queue to the display, and eliminating **multi-minute load times**

### Crum and Forster

Morristown, NJ

*Software Engineer / Data Science Intern*

*May 2025 – November 2025*

- Cut state filing prep from a **1+ hour manual process** to **under 10 minutes** by building a Python tool that runs SQL queries in parallel, computes the actuarial figures, and auto-formats a submission-ready exhibit
- Surfaced the strongest predictors of **claim frequency and severity** by building a Python (pandas, Seaborn) framework for univariate and bivariate analysis of **500K+ claim records**, informing variable selection for a downstream pricing model
- Built a monthly Python **ETL pipeline** that loads **50K+ rows** of premium and claim data into a structured **SQL Server** schema, deduplicating records, validating data quality, and normalizing column names with regex

### Launch (Startup)

Medford, MA

*Full Stack Software Engineer Intern*

*January 2025 – May 2025*

- Led a **5-intern backend team** designing a normalized **PostgreSQL** and **MongoDB** schema to model company, product, and user data
- Built a **RAG pipeline** (Ollama + scraped social data) powering an AI agent that auto-generates company websites from existing profiles

## PROJECTS

**Shipping Container Inventory Platform** | *TypeScript, React, Node.js, PostgreSQL, AWS* *May 2024 – Present*

- **Sole engineer and operator** of a live, revenue-generating inventory and invoicing platform (**40K+ lines**, **275+ tests**) for a working shipping-container yard, tracking **248 containers** and **\$657K** in active inventory
- Built an **idempotent month-end billing engine** (cron) that derives storage charges from intake/checkout dates across three billing modes, generating **line-itemized, auditable** invoices safe to re-run without double-charging
- Automated deploys with a zero-touch **GitHub Actions** pipeline that builds Docker images and ships them to EC2, plus a **transactional migration runner** that backs up, applies, and validates **25 schema migrations** atomically

**Recursive Bayesian Radar-Trace Classifier** | *Python, NumPy, Pandas*

*May 2025*

- Built a **recursive Bayesian estimator** from scratch in NumPy that classifies radar tracks as bird vs. aircraft from **600-sample velocity time series**, propagating a per-measurement posterior through a sticky-state **transition matrix** with NaN-robust updates
- Engineered a **rolling-volatility feature** with **Laplace-smoothed** empirical likelihoods from a 12,000-measurement corpus, fusing it with speed via **log-linear likelihood pooling** to lift held-out track accuracy from **9/10 to 10/10**

**JumBuddy** – *Overall Winner & Education Track, JumboHack 2026*

*February 2026*

- Architected the system and built the **capture pipeline** for a **VS Code extension**: keystroke-level edit tracking flushed to a **Flask** ingestion API, computing per-symbol metrics (thrash, churn, time-on-function) to help professors diagnose where students struggle

## TECHNICAL SKILLS

**Languages:** C++, C, Python, SQL, TypeScript, JavaScript

**Libraries & Frameworks:** NumPy, pandas, scikit-learn, Seaborn, React, Node.js, Express, Flask, FastAPI

**Tools:** Git, Docker, GitHub Actions, AWS (EC2, S3, Textract), PostgreSQL, SQL Server, MongoDB, Linux

## LEADERSHIP & ACTIVITIES

**Tufts Varsity Football:** Defensive Line

*Aug 2023 – May 2024*

**Tufts Torn Ticket II:** Stage Manager, Pit Guitarist, Actor

*Aug 2023 – Present*

**General Interests:** Guitar, Welding, Theatre, Pickleball, Chess